

Q7a) Explain Hadoop Architecture with Neat Diagram. Highlight the Roles of NameNode and DataNode.

Hadoop Architecture is a distributed framework used for storing and processing large volumes of data across multiple computers.

It mainly consists of: HDFS (Hadoop Distributed File System), MapReduce Processing Framework. Hadoop follows a Master-Slave architecture.

Hadoop Architecture Diagram -

Components of Hadoop Architecture

1. NameNode - NameNode is the master node of HDFS that manages metadata and controls file system operations.

Responsibilities of NameNode - Maintains file system namespace, Stores metadata information, Keeps track of file locations, Manages DataNodes, Handles client requests

Metadata Stored by NameNode - File names, File permissions, Block locations, Directory structure

Characteristics - Central management node, Does not store actual data, Critical component of Hadoop

Example - If a user requests a file: NameNode tells where file blocks are stored.

2. DataNode - DataNode is a slave node that stores actual data blocks.

Responsibilities of DataNode - Store data blocks, Read and write data, Send heartbeat to NameNode, Perform block replication

Characteristics - Stores actual data, Multiple DataNodes exist, Works under NameNode control

Example - A large file is divided into blocks and stored across multiple DataNodes.

Working of Hadoop Architecture -

Step 1 - Client sends request to NameNode.

Step 2 - NameNode provides block locations.

Step 3 - Client accesses DataNodes directly.

Step 4 - DataNodes store and retrieve data blocks.

Advantages of Hadoop Architecture - Distributed storage, Fault tolerance, Scalability, High availability, Parallel processing

Hadoop Architecture uses NameNode for metadata management and DataNodes for storing actual data blocks. This distributed architecture enables efficient Big Data storage and processing.

Q7b) Compare Tableau, Power BI, and Matplotlib for Data Visualization. Discuss Scenarios where Each Tool is Best Suited.

Feature	Tableau	Power BI	Matplotlib
Type	Business Intelligence Tool	Business Analytics Tool	Python Visualization Library
Developed By	Tableau Software	Microsoft	Python Community
Ease of Use	Very user-friendly	Easy to use	Requires programming
Programming Needed	No	Minimal	Yes
Best For	Interactive dashboards	Business reporting	Custom visualizations
Integration	Multiple data sources	Microsoft ecosystem	Python applications
Cost	Expensive	Affordable	Free/Open Source
Flexibility	High	Moderate	Very high

1. Tableau - Tableau is a powerful Business Intelligence and visualization tool used to create interactive dashboards and reports.

Features - Drag-and-drop interface, Real-time analytics, Interactive dashboards, Supports large datasets

Best Suited Scenarios - Business analytics, Executive dashboards, Sales analysis, Enterprise reporting

Example - A company uses Tableau to visualize yearly sales performance.

2. Power BI - Power BI is a Microsoft business analytics tool used for reporting and dashboard creation.

Features - Easy integration with Excel, Cloud-based dashboards, Real-time monitoring, Affordable solution

Best Suited Scenarios - Organizations using Microsoft products, Financial reporting, KPI dashboards, Small and medium businesses

Example - An organization uses Power BI for employee performance reporting.

3. Matplotlib - Matplotlib is a Python library used for creating static, animated, and interactive graphs.

Features - Highly customizable, Supports many graph types, Works with Python libraries like NumPy and Pandas

Best Suited Scenarios - Data Science projects, Research and analysis, Machine learning visualization, Academic work

Example - A data scientist uses Matplotlib to plot machine learning results.

Types of Graphs Supported by Matplotlib - Line charts, Bar graphs, Pie charts, Histograms, Scatter plots

Tableau is best for interactive enterprise dashboards, Power BI is ideal for Microsoft-based business reporting, and Matplotlib is suitable for custom visualizations in Data Science and machine learning projects.

Q8a) What is Data Visualization? Describe the Challenges of Data Visualization.

Data Visualization is the graphical representation of data and information using Charts, Graphs, Maps, Dashboards, Diagrams.

It helps users understand complex data easily and identify patterns, trends, and relationships.

Objectives of Data Visualization - Simplify complex data, Improve understanding, Support decision-making, Identify trends and patterns, Detect Outliers and Errors

Examples of Data Visualization - Bar chart, Pie chart, Line graph, Scatter plot, Heat map

Applications of Data Visualization - Business analytics, Healthcare reporting, Financial analysis, Scientific research, Big Data analytics

Advantages of Data Visualization - Easy interpretation of data, Faster decision-making, Identifies hidden patterns, Improves communication

Challenges of Data Visualization - Big Data contains huge amounts of structured and unstructured data, making visualization challenging.

1. Large Volume of Data - Big Data contains huge datasets which are difficult to visualize efficiently.

Problem - Graphs may become cluttered and confusing.

2. Data Quality Issues - Incorrect or incomplete data leads to misleading visualizations.

Example - Missing values affecting charts.

3. Choosing Appropriate Visualization - Selecting wrong chart type may produce incorrect interpretation.

Example - Using pie chart for large datasets.

4. Real-Time Data Handling - Visualizing continuously changing data is difficult.

Example - Stock market dashboards.

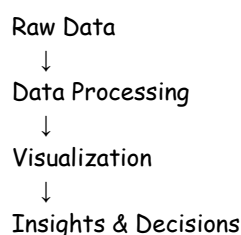
5. Data Integration - Combining data from multiple sources can be challenging.

6. Scalability - Visualization systems must handle growing datasets efficiently.

7. User Understanding - Complex visualizations may be difficult for non-technical users.

8. Performance Issues - Large visualizations may require high memory and processing power.

Diagram of Data Visualization Process -



Data Visualization transforms complex data into understandable graphical formats. However, challenges such as large data volume, scalability, and choosing proper visualization techniques must be addressed for effective analysis.

Q8b) Short Notes

i) MapReduce - MapReduce is a programming model in Hadoop used for processing large datasets in parallel across distributed systems.

It divides processing into two phases: Map Phase & Reduce Phase

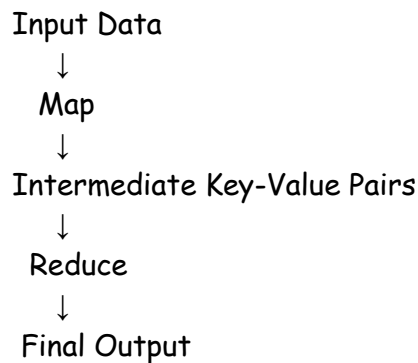
Working of MapReduce -

1. **Map Phase** - Input data is divided into smaller parts and processed in parallel.

Output - Key-value pairs are generated.

2. **Reduce Phase** - Intermediate results are combined and summarized.

MapReduce Diagram -



Advantages - Parallel processing, Fault tolerance, Scalable, Efficient for Big Data

Applications - Log analysis, Search engines, Data mining

MapReduce enables distributed processing of large datasets efficiently using Map and Reduce operations.

ii) **HDFS (Hadoop Distributed File System)** - HDFS is a distributed file system used in Hadoop to store large volumes of data across multiple machines.

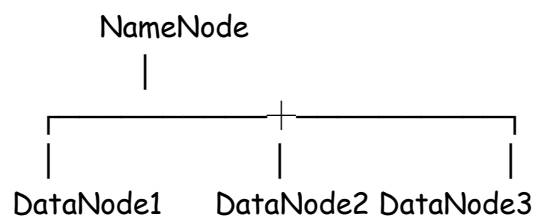
It follows Master-Slave architecture.

Components of HDFS -

1. **NameNode** - Master node, Stores metadata, Manages file system

2. **DataNode** - Slave node, Stores actual data blocks

HDFS Architecture Diagram -



Features of HDFS - Fault tolerance, Distributed storage, Scalability, High throughput

Advantages - Reliable storage, Handles Big Data efficiently, Supports parallel processing

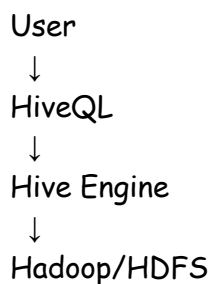
HDFS is the storage layer of Hadoop that distributes and stores data across multiple nodes efficiently.

iii) **Hive** - Hive is a Hadoop-based data warehouse tool used for querying and analyzing large datasets stored in HDFS.

It provides SQL-like language called: HiveQL

Features of Hive - SQL-like querying, Works on Hadoop, Handles large datasets, Easy data summarization

Architecture of Hive



Components of Hive - HiveQL Parser, Compiler, Metastore, Execution Engine

Advantages - Easy for SQL users, Supports Big Data analysis, Scalable, Fault tolerant

Applications - Data warehousing, Log analysis, Business intelligence, Report generation

Hive simplifies Big Data querying in Hadoop using HiveQL and provides efficient data analysis capabilities on distributed datasets.

Q8b) Write a Note on Microsoft Power BI and Qlik

1. **Microsoft Power BI** - Microsoft Power BI is a business analytics and data visualization tool developed by Microsoft.

It is used to create: Interactive dashboards, Reports, Visual analytics

Features of Power BI - Easy integration with Excel and Microsoft products, Interactive dashboards, Real-time data monitoring, Drag-and-drop interface, Cloud-based reporting

Components of Power BI -

Power BI Desktop -> Report creation, Power BI Service -> Cloud-based sharing, Power BI Mobile -> Mobile access.

Advantages - User-friendly, Affordable, Real-time analytics, Strong Microsoft integration

Applications - Business reporting, Financial analysis, KPI dashboards, Sales analytics

Example - A company uses Power BI to visualize monthly sales and employee performance reports.

Microsoft Power BI is a powerful and easy-to-use business intelligence tool for creating dashboards, reports, and real-time business analytics.

2. Qlik - Qlik Sense is a Business Intelligence (BI) and data visualization platform developed by Qlik.

It is used for: Interactive analytics, Dashboard creation, Data exploration

Features of Qlik - Associative data model, Self-service analytics, Interactive visualizations, Fast data processing, Drag-and-drop dashboard creation

Associative Model in Qlik - Qlik automatically discovers relationships among datasets.

This allows users to: Explore hidden patterns, Analyze connected data easily

Advantages - Fast analytics, Interactive dashboards, User-friendly interface, Handles large datasets efficiently

Applications - Business intelligence, Data exploration, Customer analytics, Operational reporting

Example - Retail companies use Qlik to analyze customer purchasing behavior and sales performance.

Power BI and Qlik are popular business intelligence and visualization tools. Power BI is ideal for Microsoft-based reporting environments, while Qlik is known for powerful associative analytics and interactive data exploration.

Q7a) What is a Histogram? How is it used to visualize the distribution of data? How is it different from a Density Plot?

A Histogram is a graphical representation of the frequency distribution of numerical data.

It uses: Rectangular bars, Continuous intervals called bins, to show how data values are distributed.

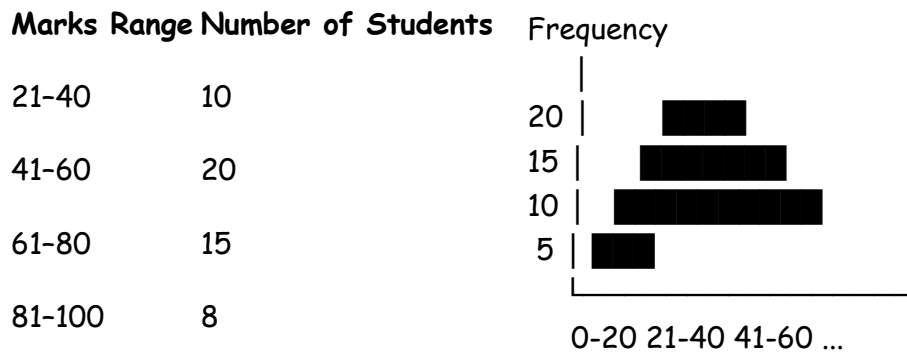
Components of Histogram - X-axis → Data intervals (bins), Y-axis → Frequency/count, Bars → Represent frequency of data.

Example of Histogram - Suppose marks of students are grouped as:

Marks Range Number of Students

0-20	5
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Histogram Diagram -



Histogram displays these frequencies using bars.

Uses of Histogram -

1. **Visualize Data Distribution** - Shows whether data is: Normal, Skewed, Uniform
2. **Identify Frequency Patterns** - Shows how often values occur.
3. **Detect Outliers** - Extreme values become visible easily.
4. **Understand Spread of Data** - Helps analyze data variation and concentration.

Density Plot - A Density Plot is a smooth curve representing the probability distribution of continuous data.

It is similar to a histogram but provides a smoother visualization.

Features of Density Plot - Uses continuous curve, Shows probability density, Better for large datasets

Difference Between Histogram and Density Plot

Histogram	Density Plot
Uses bars	Uses smooth curve
Displays frequency counts	Displays probability density
Data divided into bins	No bins required
Less smooth visualization	Smooth visualization

Easier for simple frequency analysis Better for distribution comparison

Example - Histogram shows number of students in each marks range. Density plot shows smooth probability distribution of marks.

A histogram visualizes frequency distribution using bars, while a density plot provides a smooth representation of probability distribution. Both are useful for understanding data patterns and distribution.

Q7b) What is Hadoop Ecosystem and its Primary Components? What is MapReduce and how does it fit into Hadoop Ecosystem?

The Hadoop Ecosystem is a collection of tools and frameworks that work together with Hadoop for: Big Data storage, Data processing, Data analysis, Data management

It provides distributed processing and storage capabilities.

Need for Hadoop Ecosystem - Handle massive data volumes, Distributed processing, Fault tolerance, Scalability

Primary Components of Hadoop Ecosystem -

1. **HDFS (Hadoop Distributed File System)** - Distributed storage system used to store Big Data across multiple machines.

Features - Fault tolerance, High throughput, Distributed storage

2. **MapReduce** - Programming model used for parallel data processing.

Phases - Map phase & Reduce phase

3. **YARN (Yet Another Resource Negotiator)** - Manages cluster resources and job scheduling.

4. **Hive** - Data warehouse tool used for SQL-like querying.

Language - HiveQL

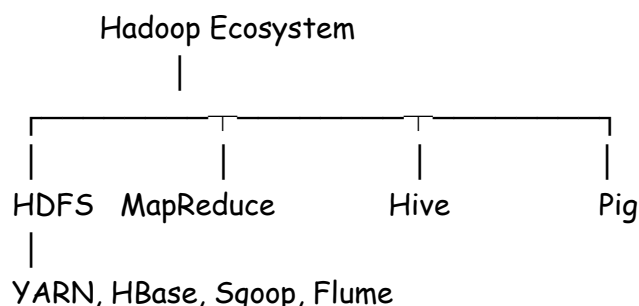
5. **Pig** - High-level scripting platform for data processing.

6. **HBase** - NoSQL distributed database for real-time data access.

7. **Sqoop** - Transfers data between Hadoop and relational databases.

8. **Flume** - Collects and transfers streaming log data into Hadoop.

Hadoop Ecosystem Diagram -



MapReduce ...

Role of MapReduce in Hadoop Ecosystem - Core processing engine of Hadoop, Processes data stored in HDFS, Performs distributed parallel computation, Handles large-scale analytics efficiently.

The Hadoop Ecosystem consists of multiple tools such as HDFS, MapReduce, Hive, Pig, and YARN for Big Data processing and storage. MapReduce acts as the core distributed processing framework that enables efficient parallel computation on large datasets.

Q8a) What is a Box Plot? Explain its Components and Interpretation.

A Box Plot (Box-and-Whisker Plot) is a graphical representation used to display the distribution, spread, and skewness of numerical data using quartiles.

It helps identify: Median, Variability, Outliers, Data spread

Components of a Box Plot - A box plot is mainly based on the five-number summary:

Minimum

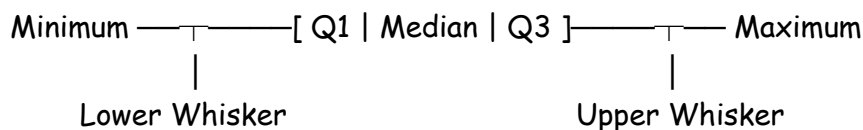
Q1 (First Quartile)

Median (Q2)

Q3 (Third Quartile)

Maximum

Structure of Box Plot



Different Components of Box Plot -

1. **Minimum Value** - Smallest data value excluding outliers.
2. **First Quartile (Q1)** - 25% of data values lie below Q1. It represents lower quartile.
3. **Median (Q2)** - Middle value of dataset. 50% of values lie below median and 50% above it.
4. **Third Quartile (Q3)** - 75% of data values lie below Q3. It represents upper quartile.
5. **Maximum Value** - Largest data value excluding outliers.
6. **Whiskers** - Lines extending from box to minimum and maximum values. They show spread of data.
7. **Outliers** - Data points far away from rest of dataset. Usually represented by dots or stars.

Interpretation of Median, Quartiles, and Whiskers -

Median - Represents center of data. If median is near center of box → symmetric distribution. If shifted → skewed distribution.

Quartiles - Q1 and Q3 divide data into four equal parts. Show spread and variability of data.

Whiskers - Represent range of data excluding outliers. Longer whisker indicates higher spread.

Interquartile Range (IQR) - IQR represents spread of middle 50% of data.

Formula - $IQR = Q3 - Q1$

Importance of IQR - Measures variability, Detects outliers, Resistant to extreme values

Example - Suppose: $Q1 = 20$ $Q3 = 50$

Then: $IQR = 50 - 20 = 30$

Advantages of Box Plot - Easy visualization of distribution, Detects outliers, Shows skewness and spread

Applications - Statistical analysis, Data preprocessing, Comparing datasets

A box plot visually summarizes data distribution using quartiles, median, whiskers, and outliers. The interquartile range (IQR) represents the spread of the middle 50% of the data.

Q8b) Explain the Role of Apache Pig in Hadoop Workflows. What is Apache Spark and how does it complement Hadoop?

Apache Pig is a high-level platform used for processing and analyzing large datasets on Hadoop.

It uses a scripting language called Pig Latin to simplify data processing tasks.

Need for Apache Pig - Writing MapReduce programs directly is complex. Pig simplifies: Data transformation, Data analysis, ETL operations

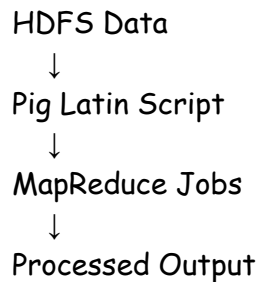
Features of Apache Pig - Easy scripting language, Runs on Hadoop, Supports structured and semi-structured data, Reduces coding complexity

Role of Apache Pig in Data Processing Workflow -

1. **Data Loading** - Loads data from HDFS into Pig environment.
2. **Data Transformation** - Performs: Filtering, Grouping, Sorting, Joining
3. **Data Processing** - Pig scripts are automatically converted into MapReduce jobs.

4. Data Analysis - Used for large-scale data analysis.

Apache Pig Workflow



Advantages of Apache Pig - Simplifies Hadoop programming, Faster development, Less code than MapReduce, Handles Big Data efficiently

Applications of Apache Pig - ETL processing, Log analysis, Data transformation, Big Data analytics

Apache Spark is a fast distributed Big Data processing framework used for: Real-time processing, Batch processing, Machine learning, Stream analytics

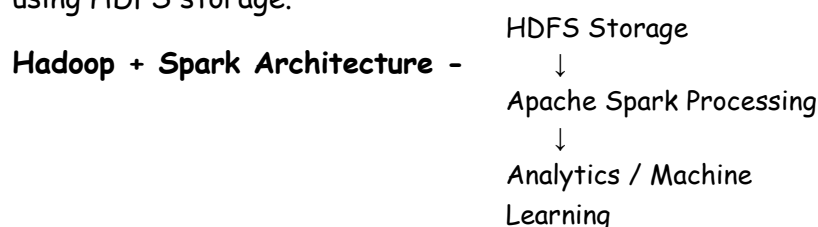
It performs in-memory computation for faster execution.

Features of Apache Spark - High-speed processing, In-memory computation, Fault tolerance, Supports multiple languages

Components of Apache Spark -	Component	Purpose
	Spark Core	Basic processing engine
	Spark SQL	SQL processing
	MLlib	Machine learning
	Spark Streaming	Real-time stream processing
	GraphX	Graph processing

Hadoop	Spark
Provides distributed storage using HDFS	Provides fast data processing
Uses disk-based MapReduce	Uses in-memory computation
Slower processing	Faster execution
Good for batch processing	Supports real-time analytics

Working Together - Hadoop HDFS stores Big Data. Spark processes data quickly using HDFS storage.



Advantages of Spark over MapReduce - Faster execution, Real-time processing, Supports machine learning,, Better iterative processing

Applications of Apache Spark - Real-time analytics, Recommendation systems, Fraud detection, Stream processing

Apache Pig simplifies Hadoop data processing using Pig Latin scripts, while Apache Spark complements Hadoop by providing fast in-memory Big Data processing and advanced analytics capabilities.

Q7a) List Few Data Visualization Tools and Discuss Any Four Applications of Data Visualization along with Various Plots

Data visualization tools are software applications used to represent data graphically.

Few Popular Data Visualization Tools -

Tool	Description
Tableau	Interactive dashboards and analytics
Microsoft Power BI	Business reporting and dashboards
Matplotlib	Python plotting library
Seaborn	Statistical data visualization
Qlik Sense	Self-service analytics
ggplot2	Data visualization in R
Plotly	Interactive web-based graphs

Applications of Data Visualization -

- 1. Business Analytics** - Visualize business performance and KPIs. Plots Used - Bar charts, Line graphs, Pie charts. Example - Monthly sales dashboard in Power BI.
- 2. Healthcare Analytics** - Analyze patient records and disease trends. Plots Used - Heat maps, Scatter plots, Histograms. Example - COVID-19 case trend visualization.
- 3. Financial Analysis** - Analyze stock market and financial performance. Plots Used - Candlestick charts, Line graphs, Area charts, Example - Stock price prediction charts.
- 4. Scientific and Research Analysis** - Represent experimental and statistical data. Plots Used - Box plots, Histograms, Scatter plots. Example - Research data distribution analysis.

Common Plots Used in Data Visualization -

1. Line Plot - Shows trends over time.

Example in Python -

```
import matplotlib.pyplot as plt
x=[1,2,3]
y=[10,20,30]
plt.plot(x,y)
plt.show()
```

2. Bar Plot - Compares categories.

Example -

```
plt.bar(["A","B","C"],[5,7,3])
plt.show()
```

3. Histogram - Shows frequency distribution.

Example -

```
plt.hist([10,20,20,30,40])
plt.show()
```

4. Scatter Plot - Shows relationship between variables.

Example -

```
plt.scatter(x,y)
plt.show()
```

5. Box Plot - Shows distribution and outliers.

Example -

```
plt.boxplot([1,2,3,4,10])
plt.show()
```

Visualization tools such as Tableau, Power BI, Matplotlib, and ggplot2 help represent complex data graphically. Different plots are used in business, healthcare, finance, and scientific applications for better analysis and decision-making.

Q7b) List the Challenges of Data Visualization. Explain the Types of Visualization with Example.

Challenges of Data Visualization ...

Types of Visualization -

1. Comparison Visualization - Compares values among categories.

Charts Used - Bar chart, Column chart

Example - Comparing sales of different products.

2. Distribution Visualization - Shows data spread and frequency.

Charts Used - Histogram, Box plot

Example - Student marks distribution.

3. Relationship Visualization - Shows relationship between variables.

Charts Used - Scatter plot, Bubble chart

Example - Relationship between height and weight.

4. Composition Visualization - Shows parts of a whole.

Charts Used - Pie chart, Stacked bar chart

Example - Market share analysis.

5. Trend Visualization - Shows changes over time.

Charts Used - Line graph, Area chart

Example - Monthly sales trends.

Data visualization faces challenges such as large data volume, scalability, and data quality issues. Different visualization types such as comparison, distribution, relationship, and trend visualization help represent data effectively for analysis and decision-making.